

Composite Trapezoidal Rule & Composite Simpson 1/3 Rule

Exam-style step-by-step derivations, error terms, and usage notes

Composite Trapezoidal Rule (Closed Newton–Cotes, $n = 1$)

Goal. Derive the composite trapezoidal formula for $\int_a^b f(x) dx$ from the closed Newton–Cotes $n = 1$ block and give the global error term.

Step 1: Single-panel (closed) Newton–Cotes for $n = 1$.

Let $x_0 = a$ and $x_1 = a + h = b$ where $h = b - a$ for the single panel. The degree-1 interpolant through $f(x_0)$ and $f(x_1)$ is the linear Lagrange polynomial $P_1(x)$. Integrating $P_1(x)$ on $[x_0, x_1]$ yields the closed Newton–Cotes block (trapezoidal single panel):

$$\int_{x_0}^{x_1} f(x) dx = \int_{x_0}^{x_1} P_1(x) dx = \frac{h}{2}(f(x_0) + f(x_1)) - \frac{h^3}{12}f''(\xi), \quad \xi \in (x_0, x_1).$$

The remainder term follows from the interpolation error for degree 1.

Step 2: Partition the interval $[a, b]$.

Divide $[a, b]$ into n equal subintervals of width

$$h = \frac{b - a}{n}, \quad x_i = a + ih, \quad i = 0, 1, \dots, n.$$

Apply the single-panel closed formula on each subinterval $[x_i, x_{i+1}]$:

$$\int_{x_i}^{x_{i+1}} f(x) dx = \frac{h}{2}(f(x_i) + f(x_{i+1})) - \frac{h^3}{12}f''(\xi_i),$$

with $\xi_i \in (x_i, x_{i+1})$ for each i .

Step 3: Sum over all subintervals (composite step).

Summing from $i = 0$ to $n - 1$,

$$\begin{aligned} \int_a^b f(x) dx &= \sum_{i=0}^{n-1} \int_{x_i}^{x_{i+1}} f(x) dx \\ &= \frac{h}{2} \sum_{i=0}^{n-1} (f(x_i) + f(x_{i+1})) - \frac{h^3}{12} \sum_{i=0}^{n-1} f''(\xi_i). \end{aligned}$$

Step 4: Combine like terms (the trapezoidal pattern).

The sum of the endpoint contributions collapses to the familiar pattern:

$$\sum_{i=0}^{n-1} (f(x_i) + f(x_{i+1})) = f(x_0) + 2 \sum_{i=1}^{n-1} f(x_i) + f(x_n).$$

Hence the composite trapezoidal rule becomes

$$\boxed{\int_a^b f(x) dx = \frac{h}{2} \left(f(x_0) + 2 \sum_{i=1}^{n-1} f(x_i) + f(x_n) \right) - \frac{h^3}{12} \sum_{i=0}^{n-1} f''(\xi_i)} \quad (1)$$

which is the exact sum of blocks with the local remainders.

Step 5: Global error via the Mean Value Theorem for sums.

Assuming f'' is continuous on $[a, b]$, by the Mean Value Theorem for integrals (applied to the sum of the local remainders) there exists $\xi \in (a, b)$ such that

$$\sum_{i=0}^{n-1} f''(\xi_i) = n f''(\xi).$$

Because $n = (b - a)/h$, we obtain the global error term

$$E_T = -\frac{(b-a)}{12}h^2 f''(\xi), \quad \xi \in (a, b), \quad (2)$$

so the composite trapezoidal rule is $\mathcal{O}(h^2)$.

Final composite trapezoidal formula (usable form).

$$\int_a^b f(x) dx \approx \frac{h}{2} \left(f(a) + 2 \sum_{i=1}^{n-1} f(a + ih) + f(b) \right), \quad E_T = -\frac{(b-a)}{12}h^2 f''(\xi).$$

When and how to use (brief).

- Use the composite trapezoidal rule when f is sufficiently smooth and you need a simple, robust integrator; it is exact for polynomials up to degree 1.
- Convergence rate: second order in h ($\mathcal{O}(h^2)$). Reduce h (increase n) to decrease error; if f'' is large or singular, error constants can be large.
- Advantage: simple implementation; useful when function values at endpoints are available.

Composite Simpson 1/3 Rule (Closed Newton–Cotes, $n = 2$)

Goal. Derive the composite Simpson 1/3 rule from the single-panel closed Newton–Cotes $n = 2$ block and give the global error term.

Step 1: Single-panel Newton–Cotes for $n = 2$ (Simpson 1/3).

Consider nodes $x_0 = a$, $x_1 = a + h$, $x_2 = a + 2h = b$, where here the single-panel spacing is $h = (b - a)/2$ for a one-panel interval $[x_0, x_2]$. Integrate the quadratic Lagrange interpolant $P_2(x)$ through $f(x_0), f(x_1), f(x_2)$ to obtain the single-panel Simpson 1/3 formula:

$$\int_{x_0}^{x_2} f(x) dx = \int_{x_0}^{x_2} P_2(x) dx = \frac{h}{3} (f(x_0) + 4f(x_1) + f(x_2)) - \frac{h^5}{90} f^{(4)}(\xi), \quad \xi \in (x_0, x_2).$$

The remainder arises from the interpolation error for degree 2.

Step 2: Partition $[a, b]$ with even n .

Divide $[a, b]$ into n equal subintervals with

$$h = \frac{b-a}{n}, \quad n \text{ even}, \quad x_i = a + ih.$$

We group the subintervals two-by-two: $[x_0, x_2], [x_2, x_4], \dots, [x_{n-2}, x_n]$.

Step 3: Apply Simpson on each pair.

On each pair $[x_{2k}, x_{2k+2}]$ (panel width $2h$):

$$\int_{x_{2k}}^{x_{2k+2}} f(x) dx = \frac{h}{3} (f(x_{2k}) + 4f(x_{2k+1}) + f(x_{2k+2})) - \frac{h^5}{90} f^{(4)}(\xi_k),$$

with $\xi_k \in (x_{2k}, x_{2k+2})$.

Step 4: Sum over all panels (composite step).

Sum $k = 0, 1, \dots, (n/2) - 1$:

$$\begin{aligned} \int_a^b f(x) dx &= \sum_{k=0}^{n/2-1} \int_{x_{2k}}^{x_{2k+2}} f(x) dx \\ &= \frac{h}{3} \sum_{k=0}^{n/2-1} (f(x_{2k}) + 4f(x_{2k+1}) + f(x_{2k+2})) - \frac{h^5}{90} \sum_{k=0}^{n/2-1} f^{(4)}(\xi_k). \end{aligned}$$

Step 5: Combine like terms (odd/even pattern).

Collecting contributions yields the standard Simpson pattern: interior odd indices get weight 4, interior even indices (except endpoints) get weight 2. Thus

$$\int_a^b f(x) dx = \frac{h}{3} \left(f(x_0) + 4 \sum_{\substack{i=1 \\ i \text{ odd}}}^{n-1} f(x_i) + 2 \sum_{\substack{i=2 \\ i \text{ even}}}^{n-2} f(x_i) + f(x_n) \right) - \frac{h^5}{90} \sum_{k=0}^{n/2-1} f^{(4)}(\xi_k) \quad (3)$$

Step 6: Global error via Mean Value Theorem for sums.

Assuming $f^{(4)}$ is continuous on $[a, b]$, there exists $\xi \in (a, b)$ such that

$$\sum_{k=0}^{n/2-1} f^{(4)}(\xi_k) = \frac{n}{2} f^{(4)}(\xi).$$

With $n = (b - a)/h$, one derives the global error term

$$E_S = -\frac{(b-a)}{180} h^4 f^{(4)}(\xi), \quad \xi \in (a, b). \quad (4)$$

Hence the composite Simpson 1/3 rule is $\mathcal{O}(h^4)$.

Final composite Simpson 1/3 (usable form).

$$\int_a^b f(x) dx \approx \frac{h}{3} \left(f(a) + 4 \sum_{\substack{i=1 \\ i \text{ odd}}}^{n-1} f(a + ih) + 2 \sum_{\substack{i=2 \\ i \text{ even}}}^{n-2} f(a + ih) + f(b) \right), \quad E_S = -\frac{(b-a)}{180} h^4 f^{(4)}(\xi).$$

When and how to use (brief).

- Use Simpson 1/3 when f is smooth (up to fourth derivative) and you can (or must) use an even number of subintervals. It is exact for polynomials up to degree 3.
- Convergence rate: fourth order in h ($\mathcal{O}(h^4)$). If high accuracy is needed and f is smooth, Simpson 1/3 is usually superior to the trapezoidal rule for the same h .
- Note: Simpson may “cancel” certain error terms (e.g. odd-power contributions) which leads to its higher accuracy relative to trapezoidal for smooth functions.

Composite Simpson 3/8 Rule & Composite Open Newton–Cotes ($n = 1$)

Exam-style step-by-step derivations, local/global remainders, usage notes

Composite Simpson 3/8 Rule (Closed Newton–Cotes, $n = 3$)

Goal. Derive the composite Simpson 3/8 rule for $\int_a^b f(x) dx$ from the single-panel Newton–Cotes block ($n = 3$) and state the global error.

Step 1: Single-panel Newton–Cotes ($n = 3$).

Consider the one-panel interval $[x_0, x_3]$ with nodes

$$x_0 = a, \quad x_1 = a + h, \quad x_2 = a + 2h, \quad x_3 = a + 3h = b,$$

where for this single panel $h = (b-a)/3$. Let $P_3(x)$ be the cubic Lagrange interpolant through $f(x_0), f(x_1), f(x_2), f(x_3)$. Integrating P_3 on $[x_0, x_3]$ yields the single-panel Simpson 3/8 block:

$$\int_{x_0}^{x_3} f(x) dx = \int_{x_0}^{x_3} P_3(x) dx = \frac{3h}{8} (f(x_0) + 3f(x_1) + 3f(x_2) + f(x_3)) - \frac{3h^5}{80} f^{(4)}(\xi),$$

for some $\xi \in (x_0, x_3)$. The remainder follows from the interpolation error for degree 3.

Step 2: Partition $[a, b]$ with n a multiple of 3.

Divide $[a, b]$ into n subintervals of width $h = (b - a)/n$, with n a multiple of 3, and nodes $x_i = a + ih$. Group subintervals three-by-three: $[x_0, x_3], [x_3, x_6], \dots, [x_{n-3}, x_n]$.

Step 3: Apply the single block on each group and sum.

On each block $[x_{3k}, x_{3k+3}]$ (width $3h$):

$$\int_{x_{3k}}^{x_{3k+3}} f(x) dx = \frac{3h}{8} (f(x_{3k}) + 3f(x_{3k+1}) + 3f(x_{3k+2}) + f(x_{3k+3})) - \frac{3h^5}{80} f^{(4)}(\xi_k),$$

$\xi_k \in (x_{3k}, x_{3k+3})$. Sum $k = 0, \dots, n/3 - 1$:

$$\int_a^b f(x) dx = \frac{3h}{8} \sum_{k=0}^{n/3-1} (f(x_{3k}) + 3f(x_{3k+1}) + 3f(x_{3k+2}) + f(x_{3k+3})) - \frac{3h^5}{80} \sum_{k=0}^{n/3-1} f^{(4)}(\xi_k).$$

Step 4: Combine like terms.

Collecting gives the repeating coefficient pattern 1, 3, 3, 2, 3, 3, 2, $\dots$, 3, 3, 1, so the compact composite form is

$$\int_a^b f(x) dx = \frac{3h}{8} \left(f(x_0) + 3 \sum_{i \equiv 1, 2 \pmod{3}} f(x_i) + 2 \sum_{\substack{i \equiv 0 \pmod{3} \\ i \neq 0, n}} f(x_i) + f(x_n) \right) - \frac{3h^5}{80} \sum_{k=0}^{n/3-1} f^{(4)}(\xi_k).$$

Step 5: Global error via Mean Value Theorem for sums.

If $f^{(4)}$ is continuous on $[a, b]$ then there exists $\xi \in (a, b)$ with $\sum_k f^{(4)}(\xi_k) = \frac{n}{3} f^{(4)}(\xi)$. Substitute $n = (b - a)/h$ to obtain the global remainder

$$E_{3/8} = -\frac{(b-a)}{80} h^4 f^{(4)}(\xi), \quad \xi \in (a, b),$$

so composite Simpson 3/8 is $\mathcal{O}(h^4)$.

Final usable form.

$$\int_a^b f(x) dx \approx \frac{3h}{8} \left(f(a) + 3 \sum_{i \equiv 1, 2 \pmod{3}} f(a + ih) + 2 \sum_{\substack{i \equiv 0 \pmod{3} \\ i \neq 0, n}} f(a + ih) + f(b) \right), \quad E_{3/8} = -\frac{(b-a)}{80} h^4 f^{(4)}(\xi).$$

When/how to use (brief).

- Use Simpson 3/8 when n is a multiple of 3 (or to cover a leftover block when combining with Simpson 1/3). It is exact for polynomials of degree ≤ 3 .
- Convergence: fourth-order in h ($\mathcal{O}(h^4)$) for smooth f .

Composite Open Newton–Cotes for $n = 1$ (open 2-point rule)

Important clarification. *Open $n = 1$ is not the midpoint rule.* The midpoint rule corresponds to open $n = 0$ (one interior point). Open $n = 1$ uses two interior points per panel and gives a different formula and error constant.

Goal. Derive the composite open Newton–Cotes formula for $n = 1$ (two interior points per panel), provide the remainder and usage notes.

Step 1: Partition and panel setup.

Divide $[a, b]$ into N equal subintervals,

$$h = \frac{b-a}{N}, \quad x_i = a + ih, \quad i = 0, \dots, N.$$

For open $n = 1$ we operate on panels of width $H = 2h$: each panel is $[x_{2k}, x_{2k+2}]$, and the two interior (open) nodes are

$$x_{2k} + \frac{H}{3} = x_{2k} + \frac{2h}{3}, \quad x_{2k} + \frac{2H}{3} = x_{2k} + \frac{4h}{3}.$$

Step 2: Single-panel open block.

On a representative panel $[A, A + H]$ with interior nodes $A + H/3$ and $A + 2H/3$, construct the degree-1 interpolant through the two interior samples. Map to $u = (x - A)/H \in [0, 1]$ and integrate the Lagrange basis. One obtains the block:

$$\int_A^{A+H} f(x) dx = \frac{H}{2} \left(f\left(A + \frac{H}{3}\right) + f\left(A + \frac{2H}{3}\right) \right) - \frac{H^3}{36} f''(\xi), \quad \xi \in (A, A + H).$$

(Verify: $\int_0^1 L_j(u) du = \frac{1}{2}$ for each interior basis; integrate the interpolation remainder to get the local $H^3/36$ factor. The sign is $-$.)

Step 3: Express in terms of partition spacing h and make composite.

Since $H = 2h$, the single-panel formula reads

$$\int_{x_{2k}}^{x_{2k+2}} f(x) dx = h \left(f\left(x_{2k} + \frac{2h}{3}\right) + f\left(x_{2k} + \frac{4h}{3}\right) \right) - \frac{(2h)^3}{36} f''(\xi_k).$$

Assuming N is even, sum over $k = 0, \dots, N/2 - 1$:

$$\boxed{\int_a^b f(x) dx = h \sum_{k=0}^{N/2-1} \left(f\left(x_{2k} + \frac{2h}{3}\right) + f\left(x_{2k} + \frac{4h}{3}\right) \right) - \frac{(2h)^3}{36} \sum_{k=0}^{N/2-1} f''(\xi_k)}.$$

Step 4: Global error (in terms of h).

Assume f'' continuous on $[a, b]$. There exists ξ with $\sum_k f''(\xi_k) = \frac{N}{2} f''(\xi)$. Using $N = (b - a)/h$ we find the total remainder

$$E_{\text{open-1}} = -\frac{(b-a)}{36} H^2 f''(\xi) = -\frac{(b-a)}{36} (2h)^2 f''(\xi) = -\frac{(b-a)}{9} h^2 f''(\xi), \quad \xi \in (a, b).$$

Hence the composite open $n = 1$ rule is $\mathcal{O}(h^2)$.

Final usable form.

$$\boxed{\int_a^b f(x) dx \approx h \sum_{k=0}^{N/2-1} \left(f\left(a + 2kh + \frac{2h}{3}\right) + f\left(a + 2kh + \frac{4h}{3}\right) \right), \quad E = -\frac{(b-a)}{9} h^2 f''(\xi)}.$$

When/how to use (brief).

- Use open $n = 1$ when endpoint values are unavailable or unreliable (e.g. endpoint singularities) but interior samples are available.
- Accuracy: second-order in h ($\mathcal{O}(h^2)$); exact for linear polynomials. It is different from the midpoint rule (open $n = 0$) and has a different constant in the remainder.